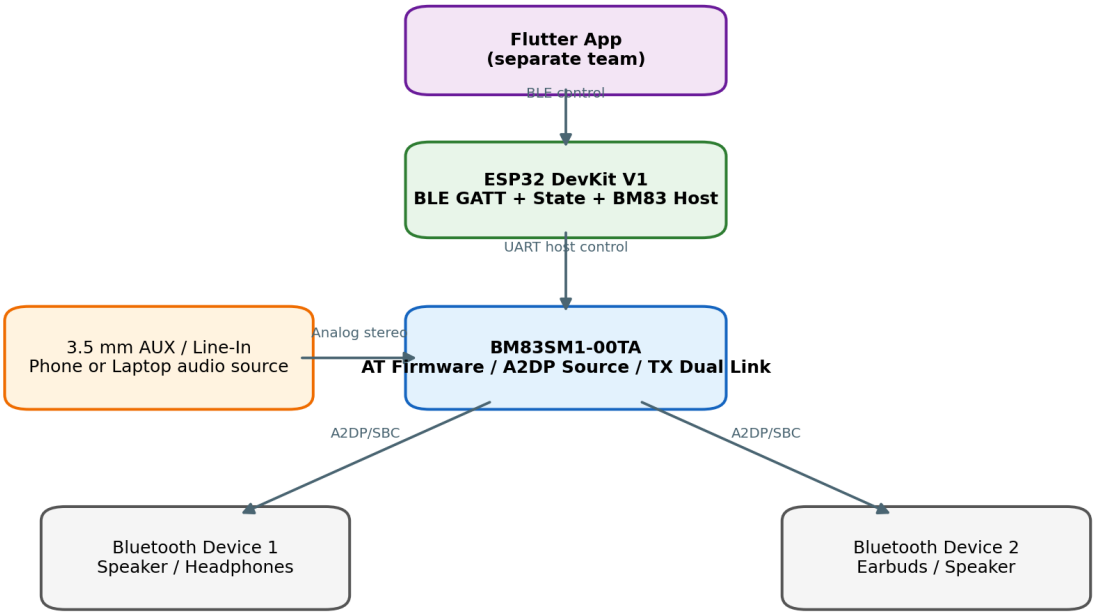
BT SHARE

Multi-Bluetooth Audio Hub

BUSINESS REQUIREMENTS DOCUMENT (BRD)

Hardware & Embedded Prototype - Two-Device Target



Audio path never goes through the Flutter phone. Control and audio are separated.

Hardware-first execution baseline | LKR 40,000 prototype ceiling | Flutter team developed separately and integrated after hardware proof

Document status: Hardware Team Working Baseline | Date: 06 Oct 2026

1. Executive Summary

BT Share is a proof-of-concept embedded Bluetooth audio hub. The hardware team shall build a physical prototype that accepts one stereo AUX/line-level source and transmits the same audio program to two independent Bluetooth Classic A2DP sink devices simultaneously. The prototype shall use the BM83 as the Bluetooth audio engine and the ESP32 as the control, state-management and mobile-integration controller.

The hardware team is required to reach a complete two-device working state before Flutter integration. The Flutter application is developed by a separate team. During final integration, the Flutter app will control the ESP32 over BLE GATT; the ESP32 will translate app actions into BM83 host commands over UART. The phone is a control interface only and is not part of the audio path.

Primary success statement

One AUX source is heard on Bluetooth Device 1 and Bluetooth Device 2 at the same time, while the system can discover, pair, connect, disconnect and report state reliably. Flutter integration is a later layer; it must not be required to prove the core two-device hardware.

2. Purpose, Audience and Document Hierarchy

Item	Definition
Purpose	Define exactly what the hardware/embedded team must deliver, the requirements that are mandatory, the integration boundary with Flutter, and the objective acceptance criteria.
Primary audience	Hardware/embedded team, project lead, test/review team, and the Flutter team at the integration boundary.
BRD role	This document defines WHAT must be achieved and accepted. Exact schematics, PCB layout details and BM83 command syntax belong in controlled engineering design documents.
Execution source	Multi_Bluetooth_Audio_Hub_BM83_0_to_100_Master_Plan - build order, gates and workflow.
Procurement source	Multi_Bluetooth_Audio_Hub_BM83_Final_Company_Procurement_Submission - approved one-shot parts/tools/fabrication envelope.
Original architecture source	Multi-Bluetooth Audio Hub Flutter Control MD - original control/audio separation, two-device milestone, device management and acceptance concepts.

Controlled planning rule

If this BRD and an older draft conflict, the latest approved BRD + Master Plan + Final Company Procurement Submission form the current prototype baseline. Changes after approval must be recorded as a requirement change, not silently implemented.

3. Business Problem and Prototype Objectives

3.1 Problem Statement

Conventional Bluetooth audio use often targets a single active receiver from a source device. The project seeks to demonstrate a dedicated external hub that can take one analog stereo source and deliver the same program to two Bluetooth audio receivers at the same time, while device selection and connection management are controlled independently from the audio path.

3.2 Prototype Objectives

- Prove two simultaneous Bluetooth Classic A2DP sink connections from one BM83-based transmitter architecture.
- Prove stable stereo AUX input and same-source audio delivery to both receivers.
- Build a deterministic ESP32 host controller for BM83 discovery, pairing, connection, state and error management.
- Expose a clean BLE GATT app-facing contract on the ESP32 without exposing BM83 internals to the Flutter team.
- Integrate the separately developed Flutter app only after the hardware subsystem works independently.
- Complete the prototype within the approved LKR 40,000 procurement ceiling and document its actual limitations.

4. Scope Boundaries

4.1 In Scope - Hardware & Embedded Team

- BM83 carrier-board schematic, PCB layout, assembly-ready documentation and bring-up.
- BM83 power, AUX input, UART host interface, reset/MFB/configuration access and test points.
- One-device A2DP audio proof followed by two-device Dual-Link proof.
- ESP32 firmware for BM83 host communication, discovery, pairing, connection, state, retry/recovery, logging and BLE GATT.
- Hardware-facing integration contract for Flutter: commands, events, identifiers, states and error codes.
- Compatibility, reconnect, range, long-run, latency-observation and final-demo qualification.
- Test evidence and final prototype handover documentation.

4.2 Out of Scope for This Prototype

- 3-5 simultaneous Bluetooth audio receivers.
- Auracast, LE Audio or BLE Audio Broadcast.
- Perfect receiver synchronization or zero-latency claims.
- Production enclosure, battery product design or production certification.
- Final mass-production RF optimization or market-ready PCB certification.
- Flutter UI implementation itself; that belongs to the separate Flutter team.
- Routing or relaying the audio stream through the Flutter phone.

5. Fixed System Architecture

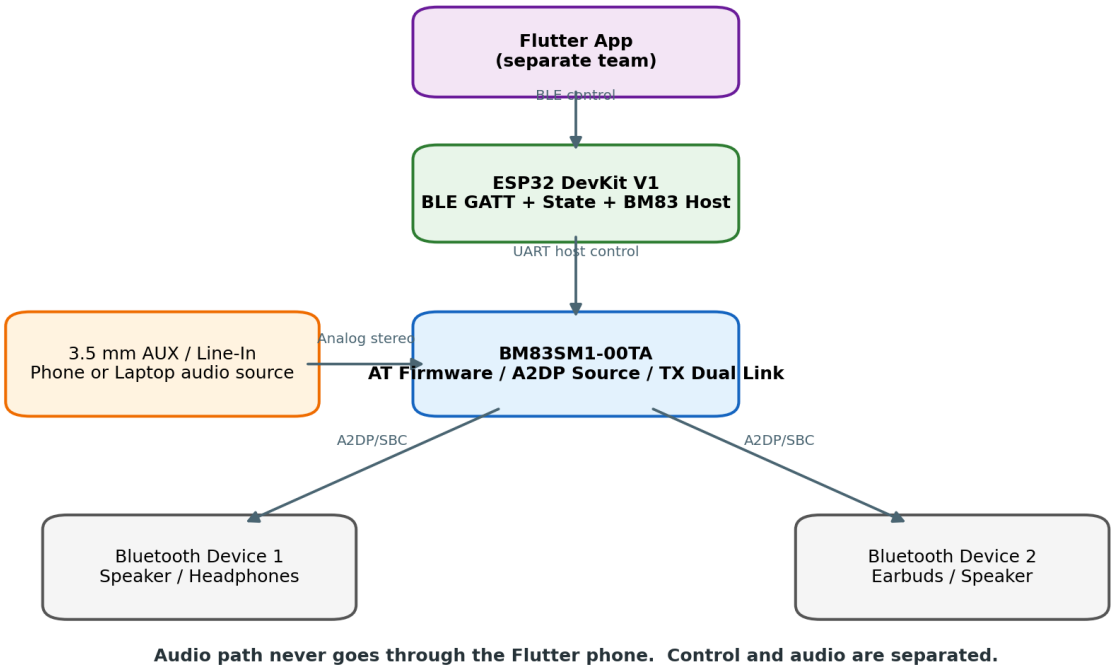


Figure 1 - Fixed prototype architecture and separation of control path from audio path.

Layer	Responsibility
Flutter App	User-facing control/status only. Connects to ESP32 over BLE GATT.
ESP32	BLE GATT server, command handler, device/state manager, BM83 host controller, logging and integration boundary.
BM83	Bluetooth Classic audio engine, discovery/pairing/connection under selected AT firmware, A2DP Source, TX mode, SBC, Dual Link and AUX audio processing.
AUX Input	External stereo line-level source. Must not accept high-power speaker output.
Bluetooth Device 1 / 2	Independent A2DP sink devices such as speaker, headphone or earbuds.

6. Constraints and Non-Negotiable Decisions

ID	Constraint / Decision	Rationale
C-001	First prototype target is exactly 2 simultaneous receivers.	Prevents scope creep and aligns development/test target.
C-002	Total prototype procurement ceiling: LKR 40,000.	Approved project constraint.

ID	Constraint / Decision	Rationale
C-003	BM83SM1-00TA is the selected Bluetooth audio engine; one active unit + one spare in procurement.	Protects schedule against single imported-module failure.
C-004	Official BM83 evaluation board is excluded. Team-designed minimal carrier PCB is used.	Budget control and project ownership.
C-005	Flutter team develops separately.	Allows parallel work and independent hardware debugging.
C-006	Core hardware must work without Flutter.	Prevents UI/BLE issues from masking audio/Bluetooth faults.
C-007	No display, rotary encoder or app-replacement physical UI.	Mobile app is the intended user interface.
C-008	Exact BM83 pin, voltage and AT-command details must be taken from the approved firmware/hardware documentation; no guessing.	Protects hardware and prevents protocol errors.
C-009	Two receivers may have different audible latency.	Dual connection does not equal perfect synchronization.
C-010	Flutter phone is never in the audio data path.	Maintains architecture separation and demo clarity.

7. Team Responsibilities and Ownership

Workstream	Hardware / Embedded Team	Flutter Team
Physical hardware	Owns BM83 carrier, power, AUX, ESP32, wiring, test points and prototype assembly.	No responsibility.
Bluetooth audio	Owns BM83 configuration, A2DP source behavior and two-receiver proof.	No responsibility.
BM83 host control	Owns UART protocol, parser, commands/events and device state.	No responsibility.
BLE transport	Owns ESP32 GATT server, UUIDs, protocol version, notifications and error model.	Owns BLE client implementation.
User functions	Implements firmware endpoints required by the frozen integration contract.	Defines final app function catalogue/UI actions and maps them to the contract.
App UI/UX	Provides hardware status semantics only.	Owns screens, navigation, UX and presentation.
Integration	Provides a hardware system already working without Flutter.	Provides app already working against mock/test BLE contract.

Workstream	Hardware / Embedded Team	Flutter Team
Acceptance	Joint integration test after each subsystem passes its own gate.	Joint integration test.

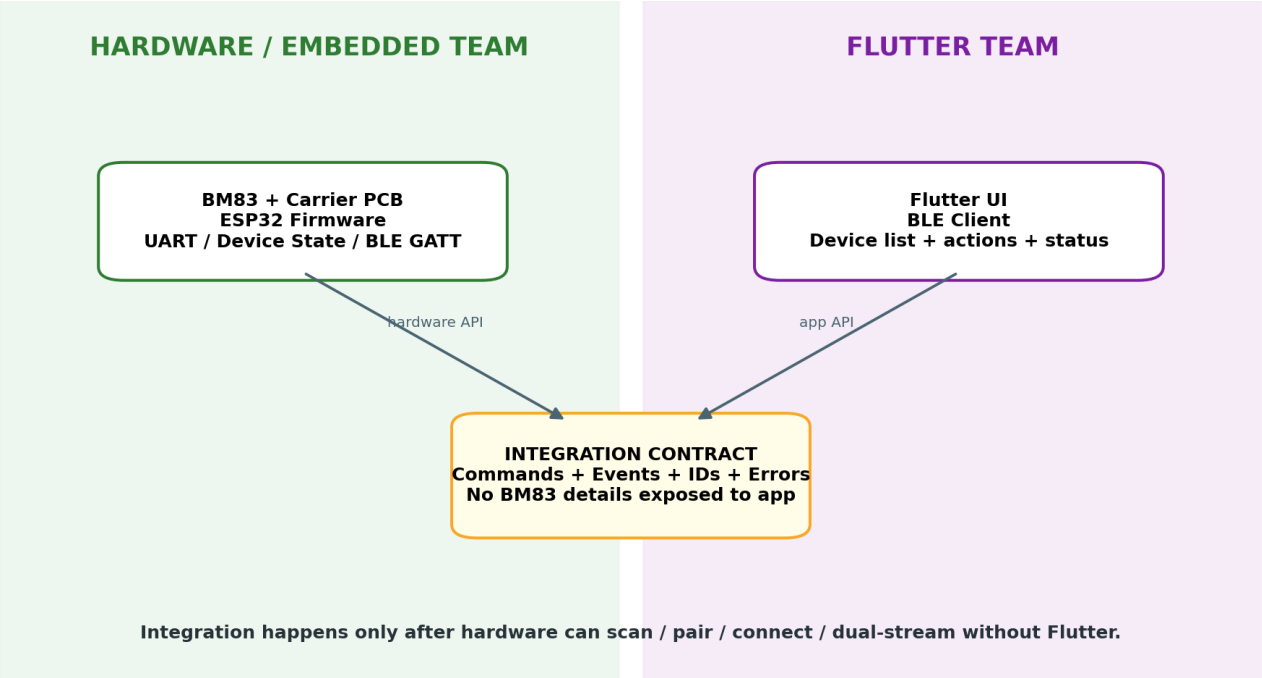


Figure 2 - Team boundary and integration contract.

8. Business Requirements

ID	Requirement	Priority
BR-001	The prototype shall demonstrate one AUX source playing through two Bluetooth A2DP sink devices simultaneously.	Must
BR-002	The hardware subsystem shall reach the two-device target before Flutter integration.	Must
BR-003	The prototype shall remain within the approved LKR 40,000 procurement ceiling.	Must
BR-004	The system shall separate control traffic from audio traffic.	Must
BR-005	The app-facing interface shall hide BM83-specific command syntax and expose stable product-level actions/states.	Must
BR-006	The system shall provide enough diagnostics to identify whether failures occur in power, UART, Bluetooth, BLE transport or app/state synchronization.	Must

ID	Requirement	Priority
BR-007	The final demo shall clearly show Device 1, Device 2 and simultaneous playback from the same source.	Must
BR-008	Limitations including latency/synchronization and compatibility shall be documented rather than overclaimed.	Must

9. Functional Requirements - Hardware and Audio

ID	Requirement	Verification
HW-001	The carrier board shall expose safe power input, ground, UART, AUX L/R, reset/MFB/control access and required test points.	Design review + inspection
HW-002	The BM83 power architecture shall comply with the approved device requirements and shall not apply unsupported voltage directly to the raw module.	Power pre-check
HW-003	The stereo AUX path shall accept a normal line-level source and preserve Left/Right channels to BM83.	Audio bench test
HW-004	The board layout shall preserve the BM83 antenna keep-out/orientation requirements from the approved hardware guidance.	PCB review
HW-005	The carrier shall provide accessible debug/test points for the main power rail, ground, UART and audio signals.	Visual inspection
AUD-001	BM83 shall operate as the A2DP Source for the prototype.	One-device audio test
AUD-002	The source audio shall be encoded/transmitted using the baseline SBC path supported by the selected setup.	Configuration record + playback
AUD-003	One known-good A2DP sink shall receive stable audio before any dual-link testing starts.	Gate test
AUD-004	Two A2DP sinks shall receive the same source program simultaneously during the core proof.	Dual-link test
AUD-005	Disconnecting one sink shall not create an undocumented failure state for the other; behavior shall be observed and recorded.	Failure injection

ID	Requirement	Verification
AUD-006	High-power speaker output shall not be used as AUX/line input.	Build/test checklist

10. Functional Requirements - Bluetooth, ESP32 and Firmware

ID	Requirement
BT-001	The system shall support discovery of compatible nearby Bluetooth audio receivers using the selected BM83 firmware path.
BT-002	The system shall distinguish pairing from active connection and maintain both states correctly.
BT-003	The system shall support connection of Device 1 and Device 2 and enforce a maximum-two prototype rule.
BT-004	The system shall detect and report disconnects and expose a defined reconnect action/policy.
FW-001	ESP32 shall communicate with BM83 over a verified UART electrical/protocol interface.
FW-002	BM83 responses/events shall be parsed into a product-level device/state model rather than passed raw to the Flutter app.
FW-003	The ESP32 shall maintain the authoritative discovered/selected/paired/connected/streaming state for each device.
FW-004	Firmware shall use explicit states: UNKNOWN, DISCOVERED, SELECTED, PAIRING, PAIRED, CONNECTING, CONNECTED, STREAMING, DISCONNECTED and FAILED/RETRY as applicable.
FW-005	Firmware shall implement command validation, timeouts, structured errors and structured serial logs.
FW-006	The system shall recover to a known state after expected connection failure/retry paths; uncontrolled silent state loss is not acceptable.
FW-007	Core device discovery/pair/connect/disconnect/dual-stream operation shall be testable from serial/diagnostic control before BLE/Flutter integration.
FW-008	BLE control activity shall not interrupt normal A2DP audio streaming under normal prototype operation.

11. Flutter Integration Requirements and Function Handover

Important dependency
The Flutter team has not yet issued the final function catalogue. Hardware firmware must therefore be architected around a stable product-level command/event contract and mockable handlers. When Flutter provides its final list, the hardware team maps the approved functions to these handlers without changing the audio architecture.

11.1 Provisional Minimum App Function Baseline

ID	Function	Hardware expectation
APP-F01	Connect to BT Share hub	BLE connection to ESP32; show hub ready/not ready.
APP-F02	Start receiver scan	Request BM83 discovery through ESP32.
APP-F03	Display discovered devices	Show name/identifier and current state from ESP32.
APP-F04	Select Device 1	Assign one discovered device to slot 1.
APP-F05	Select Device 2	Assign one discovered device to slot 2; reject third active selection.
APP-F06	Pair device	Send pair request and show progress/result.
APP-F07	Connect device	Connect selected/paired receiver and show state.
APP-F08	Disconnect device	Disconnect one selected receiver and update state.
APP-F09	Reconnect request	Request reconnect for an eligible disconnected receiver.
APP-F10	View status	Show connected/disconnected/streaming states for both slots.
APP-F11	View errors	Display clear pairing/connection/max-link/BM83 communication errors.
APP-F12	Development log/status view	Optional development screen for protocol/status diagnostics during integration.

11.2 Flutter Team Function Catalogue Required Before Integration Freeze

Flutter team shall return a function list using the following fields. Hardware shall not guess UI behavior that is not supplied.

Field	What Flutter team must provide
Function ID	Stable ID/name for each user action.

Field	What Flutter team must provide
UI action	What the user taps/selects/requests.
Command intent	What hardware action is expected.
Parameters	Device ID, slot number or other required data.
Expected response	Immediate acknowledgement and/or final event.
Status shown	What state the app needs to display.
Error cases	What errors must be surfaced to the user.
Priority	Must / Should / Optional for first prototype.
Screen owner	Which Flutter screen/function consumes it.

11.3 Draft Hardware-to-App Event Model

ID	Event	Meaning
EVT-01	HUB_READY	ESP32 + BM83 control stack ready.
EVT-02	SCAN_STARTED	Discovery accepted/started.
EVT-03	DEVICE_FOUND	One device record added/updated.
EVT-04	SCAN_COMPLETE	Discovery cycle ended.
EVT-05	PAIRING	Pairing in progress.
EVT-06	PAIRED	Trusted relationship established.
EVT-07	CONNECTING	Profile connection in progress.
EVT-08	CONNECTED	Receiver connected.
EVT-09	STREAMING	Receiver link is in streaming state where supported by status model.
EVT-10	DISCONNECTED	Receiver link ended.
EVT-11	MAX_LINKS_REACHED	Two-device prototype limit prevents another active link/selection.
EVT-12	ERROR	Structured error code + context supplied.

12. Device and State Data Model

Field	Required meaning
deviceId / address	Stable identifier used by hardware and app contract.

Field	Required meaning
name	Human-readable Bluetooth device name where available.
paired	Trusted relationship status.
selectedSlot	None, Device 1 or Device 2.
connected	Active Bluetooth profile link status.
connectionState	Product-level state such as DISCOVERED/PAIRING/CONNECTED/STREAMING.
lastSeen	Timestamp/counter useful for scan list maintenance.
errorCode	Structured error when the latest operation fails.

State ownership rule: ESP32 is the source of truth for the product-level device list and connection state presented to Flutter. Flutter displays/sends user intent; it does not independently invent connection state.

13. Non-Functional Requirements

ID	Category	Requirement
NFR-001	Reliability	Prototype must complete repeated connect/disconnect cycles without uncontrolled ESP32/BM83 resets.
NFR-002	Diagnostics	Every major command and state transition shall be visible in structured serial logs during development.
NFR-003	Recoverability	Disconnect/out-of-range/failed connection shall result in an explicit recoverable state, not a frozen UI/system.
NFR-004	Maintainability	Firmware shall separate BLE transport, app protocol, state manager and BM83 driver/protocol responsibilities.
NFR-005	Testability	Core hardware functions shall be testable without Flutter.
NFR-006	Budget	Procurement shall not exceed LKR 40,000 without re-approval.
NFR-007	Safety	Low-voltage electronics work shall follow the approved lab/company safety process; powered/hot-tool work is supervised.
NFR-008	Transparency	Latency/synchronization and compatibility limitations shall be measured/observed and reported honestly.
NFR-009	Scope control	No 3-5 device, Auracast, display or production-product work may delay the two-device prototype.

ID	Category	Requirement
NFR-010	Integration stability	BLE command/status handling shall not intentionally route or process the audio stream on ESP32/Flutter.

14. Requirement-Driven Execution Plan

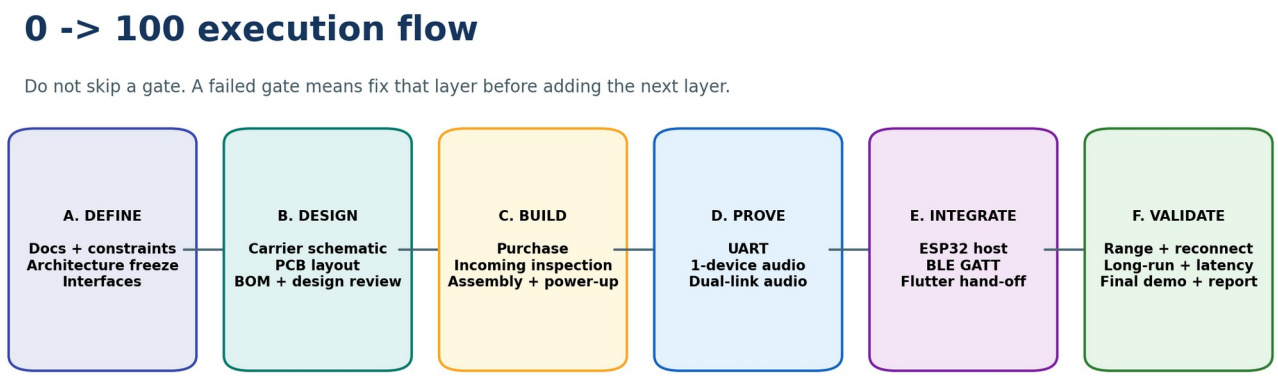


Figure 3 - Approved 0-to-100 execution flow. A failed gate is fixed before the next layer is added.

Phase	Work	Exit gate
P0 - Define	Freeze BRD, constraints, interfaces and reference docs.	BRD accepted; no architecture drift.
P1 - Design	Carrier schematic, exact BOM mapping, PCB layout, RF review, Gerbers.	Design review + ERC/DRC pass.
P2 - Build	Receive parts, inspect, assemble carrier, electrical pre-check.	Safe/stable power rails; no short.
P3 - BM83 Bring-up	UART response, firmware/configuration verification, Tx/Host/AUX setup.	BM83 alive and controllable.
P4 - One Device	AUX -> BM83 -> one receiver.	Stable audible playback.
P5 - Two Device Core Proof	AUX -> BM83 -> Device 1 + Device 2.	Same source heard on both.
P6 - ESP32 Host	UART driver/protocol, state machine, discovery/pair/connect/disconnect.	Full serial-controlled hardware flow.

Phase	Work	Exit gate
P7 - BLE GATT	Freeze product-level BLE contract and test with generic client.	BLE commands/events stable.
P8 - Flutter Integration	Map Flutter functions to frozen contract.	App can control two-device flow.
P9 - Validate	Compatibility, reconnect, range, long-run, latency observation and final demo.	Acceptance matrix passed/documented.

15. Pre-Arrival Work - Approx. 10-Day Delivery Window

The team shall not wait for physical parts. The purpose of this window is to finish everything that can be completed without the hardware so arrival triggers assembly and bring-up immediately.

Day	Mandatory output
Day 1	Archive official BM83 docs; create pin/function and firmware capability sheets; close open assumptions.
Day 2	Freeze carrier interfaces and power/AUX/UART/reset/test-point requirements.
Day 3	Complete carrier schematic V1; perform net-by-net review.
Day 4	Freeze schematic BOM mapping to the already approved procurement stock; peer review.
Day 5	Complete PCB layout, antenna keep-out review, DRC and Gerber release.
Day 6	Create ESP32 firmware repository, logger, state model, command shell and BM83 driver interfaces.
Day 7	Build mock BM83 response parser/tests and command timeout/error framework.
Day 8	Define BLE UUID/protocol draft, events and Flutter handover template.
Day 9	Prepare bring-up, one-device, dual-link, reconnect, range and long-run test sheets.
Day 10	Final cross-check: schematic/PCB/firmware/protocol/test documents ready; receiving checklist ready.

Expected readiness before parts arrive

Carrier schematic reviewed; PCB Gerbers ready/released; firmware project compiles; mock UART parser works; BLE draft contract issued; test sheets and receiving checklist prepared.

16. Procurement and Hardware Baseline

The detailed purchase links/prices are controlled in the Final Company Procurement Submission. This BRD defines only the required hardware classes and ownership relationship.

Item	Prototype role	Status
BM83SM1-00TA x2	One operational Bluetooth audio engine + one imported spare.	Approved procurement baseline
ESP32 DevKit / ESP-WROOM-32 with CP2102	Main controller and BLE GATT server.	Approved procurement baseline
Team-designed BM83 Carrier PCB	Replaces the expensive official BM83 EVB.	Engineering design deliverable
3.5 mm stereo AUX interface	External line-level audio input.	Required
Power/regulation/passives/connectors/test points	Safe BM83 carrier operation and debugging.	Required
USB-UART/debug interfaces	BM83/ESP32 bring-up and diagnostics.	Required
Bluetooth test receivers	At least two independent compatible sinks; mixed types/brands preferred.	Team/company-owned where possible
Multimeter/soldering/prototyping tools	Build and verification tools.	Procurement/company lab baseline

Procurement control
The detailed procurement document remains the price/seller/quantity authority. This BRD must not be used as a substitute for the company purchase pack.

17. Verification and Test Requirements

Test	Scenario	Pass evidence	Covers
T-01	Carrier electrical pre-check	No unintended shorts; expected low-voltage rails available before BM83 operation.	HW-001/002
T-02	BM83 UART bring-up	Valid BM83 response/activity under approved setup.	FW-001
T-03	Single sink playback	AUX -> BM83 -> one speaker/headphone produces stable audio.	AUD-001/003
T-04	Dual-link playback	Same source program audible on two independent sinks simultaneously.	BR-001, AUD-004
T-05	Discovery	Compatible receiver appears in ESP32 product device list.	BT-001, FW-002/003

Test	Scenario	Pass evidence	Covers
T-06	Pairing/connection	Pair and connect flows transition through explicit states.	BT-002/003
T-07	Disconnect/reconnect	One link drops/reconnects and state/errors remain correct.	BT-004, NFR-003
T-08	Maximum-link behavior	Third active selection/link request returns max-link state/error.	BT-003
T-09	BLE generic client	Scan/select/pair/connect/status commands work through BLE without Flutter.	BR-002, FW-007
T-10	Flutter integration	Flutter performs approved functions against same frozen contract.	BR-005
T-11	Mixed devices	Speaker+headphone, speaker+earbuds or equivalent mixed pair works/documented.	BR-008
T-12	Range	Record actual 1m/3m/5m/practical-room behavior.	NFR-008
T-13	Long run	30 min, 1 h and 2 h streaming observations recorded.	NFR-001
T-14	Power cycle/recovery	System returns to a known state after defined restart/recovery tests.	NFR-003
T-15	Latency observation	Difference between receiver outputs measured/observed and documented; no perfect-sync claim.	BR-008

18. Acceptance Criteria - Hardware Team Definition of DONE

ID	Acceptance criterion
AC-01	BM83 carrier powers reliably and BM83/ESP32 are stable.
AC-02	AUX stereo input is functional.
AC-03	One compatible A2DP sink can be discovered, paired, connected and streamed.
AC-04	Two compatible A2DP sinks can be connected simultaneously and both receive the same source program.
AC-05	ESP32 controls BM83 through UART and maintains correct product-level device/state information.
AC-06	Serial diagnostics can execute the full core flow without Flutter.

ID	Acceptance criterion
AC-07	ESP32 BLE GATT can execute the approved control/status contract with a generic BLE test client.
AC-08	Flutter integration can scan/select/pair/connect/disconnect/reconnect and display status/errors using the frozen contract.
AC-09	Disconnect/reconnect and max-link behaviors are explicit and documented.
AC-10	No unacceptable continuous audio dropouts occur during the agreed demo test.
AC-11	Range, long-run and latency/synchronization observations are documented.
AC-12	All final design, firmware, interface and test evidence is archived for handover.

19. Risk Register and Required Mitigation

ID	Risk	Impact	Mitigation
R-01	Wrong/incompatible BM83 firmware/configuration	High	Verify exact module/firmware behavior before depending on commands or Dual Link; keep configuration record.
R-02	Carrier PCB design error	High	Net-by-net review, official footprint/pin references, ERC/DRC, accessible test points and spare PCBs.
R-03	BM83 module damaged/DOA	High	Procure one spare BM83; inspect and follow approved low-voltage bring-up process.
R-04	UART electrical/protocol mismatch	High	Verify voltage/interface and protocol details before connection; debug with structured logs.
R-05	One-device audio not stable	High	Do not attempt dual-link/Flutter; isolate power, AUX, A2DP, pairing and configuration first.
R-06	Two receivers have audible delay difference	Medium	Document as expected limitation; do not claim synchronization.
R-07	Flutter requirements arrive late/change	Medium	Use provisional product-level handlers; require final function catalogue before integration freeze.

ID	Risk	Impact	Mitigation
R-08	Budget overrun	Medium	Procurement ceiling is binding; re-approval required before exceeding LKR 40,000.
R-09	Integration failure across many layers	Medium	Classify failure as BLE transport, protocol, state synchronization or BM83/hardware; debug one layer at a time.
R-10	Demo device incompatibility	Medium	Qualify a known-good primary pair and backup pair before demo day.

20. Required Hardware-Team Deliverables

ID	Deliverable	Minimum content
D-01	Approved Hardware & Embedded BRD	This document.
D-02	BM83 Firmware/Capability Study	Exact firmware/configuration, required commands/events and open/closed assumptions.
D-03	Carrier Schematic + Design Review	Power, AUX, UART, reset/MFB/config/recovery and test points.
D-04	Carrier PCB Package	PCB layout, Gerbers, fabrication notes and revision ID.
D-05	ESP32 Firmware Architecture	Modules, state machine, BM83 driver/protocol, BLE, storage/logging.
D-06	BM83 Command/Event Mapping	Product operations mapped to exact verified BM83 commands/events.
D-07	BLE Integration Contract	UUIDs, protocol version, commands, events, data model and error codes.
D-08	Flutter Function Mapping	Final Flutter team catalogue mapped to hardware endpoints.
D-09	Single-Device Test Report	Audio/connection/stability evidence.
D-10	Dual-Device Test Report	Same-source two-receiver proof and disconnect/reconnect evidence.
D-11	Compatibility/Range/Long-Run/Latency Report	Final validation results and known limitations.
D-12	Final Prototype Test & Demo Report	Acceptance matrix, final setup and demo evidence.

21. Requirements Traceability Matrix

Business Req	Derived requirement	Verification	Evidence
BR-001	AUD-003/004, BT-003	T-03/T-04	D-09/D-10
BR-002	FW-007	T-09	D-05/D-07
BR-003	NFR-006	Procurement review	Final procurement submission
BR-004	FW-002, NFR-010	Architecture review + T-10	D-05/D-07
BR-005	FW-002, APP-F01..12	T-09/T-10	D-07/D-08
BR-006	FW-005, NFR-002	Log review during all tests	D-05/D-12
BR-007	AUD-004, AC-04	Final demo	D-12
BR-008	NFR-008	T-11/T-12/T-15	D-11/D-12

22. Change Control and Integration Freeze Rules

- Architecture-changing requests require project-lead approval and BRD update before implementation.
- Flutter UI changes that do not alter hardware commands/events may proceed without hardware architecture changes.
- New Flutter functions that require new hardware behavior must be added to the function catalogue and mapped to a requirement ID before firmware work starts.
- Exact BM83 commands/packet details may be refined during firmware study without changing the product-level app contract, provided behavior remains equivalent.
- No new feature may delay AC-04: two simultaneous receivers playing the same AUX source.
- After BLE integration contract freeze, UUID/protocol breaking changes require coordinated Flutter + hardware sign-off.

23. Final Demonstration Flow

1. Power the approved prototype and allow ESP32/BM83 initialization to reach HUB_READY.
2. Connect the Flutter app to the ESP32 over BLE.
3. Start a receiver scan and display discovered compatible devices.
4. Select Device 1 and Device 2.
5. Pair/connect both receivers through the app-controlled hardware path.
6. Connect/play a known-good stereo AUX source.
7. Demonstrate audible playback on Device 1.
8. Demonstrate audible playback on Device 2.
9. Demonstrate both receiving the same source program simultaneously.
10. Optionally perform one controlled disconnect/reconnect if the qualified demo script includes it.
11. Show status/error handling and state updates in the app.
12. State documented prototype limitations honestly: latency difference, compatibility variation and two-device scope.

Demo success
The final demo is successful when the two-device audio proof and app control/status proof both work, while the audio path remains AUX -> BM83 -> Bluetooth receivers.

24. Hardware Team Start Checklist

Start item	Status
Architecture frozen to BM83 + ESP32 + AUX + two sinks	[ ]
Final procurement submission approved/ordered	[ ]
Official BM83 document pack archived	[ ]
Carrier schematic owner assigned	[ ]
PCB review owner assigned	[ ]
ESP32 firmware repository created	[ ]
Mock BM83 parser/state model started	[ ]
BLE integration contract owner assigned	[ ]
Flutter team requested to return final function catalogue	[ ]
Test templates created	[ ]
Known-good two-receiver demo candidates identified	[ ]

25. Final BRD Statement

This BRD authorizes the hardware/embedded team to proceed with the two-device BT Share prototype using the current BM83 + ESP32 execution baseline. The team shall design and prove the hardware first, complete the ESP32 host/control layer second, expose a stable BLE contract third, and integrate the separately developed Flutter application only after the standalone hardware subsystem is already functional.

The single most important product requirement is unchanged throughout the project: one external AUX source must be transmitted through the BM83 to two independent Bluetooth A2DP receivers simultaneously. All planning, procurement, firmware, testing and integration work shall support this requirement and shall not be diverted by future-product features.

Approval	Name	Signature / Date
Project Lead		
Hardware / Embedded Lead		
Flutter / Integration Representative		
Reviewer / Mentor		